

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460043.6535 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.194 +/- 0.057
Transit depth (Rp/Rs)^2	3.78 +/- 2.23 [%]
Orbital Inclination (inc)	86.51 +/- 2.91
Airmass coefficient 1 (a1)	0.099 +/- 0.063
Airmass coefficient 2 (a2)	1.71 +/- 1.6
Scatter in the residuals of the lightcurve fit is	62.97 %
Best Comparison Star	#3 - [104, 180]
Optimal Aperture	1.12
Optimal Annulus	3.04
Transit Duration (day)	0.0511 +/- 0.0036

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.194 ± 0.057 vs 0.1594 (deviation 0.6σ)
Duration fit vs published	0.0511 d vs 0.0512 d (deviation 0.04σ)
Mid-time O-C	5.4 min
Depth SNR	3.3
Residual scatter	62.97 %

Light curve

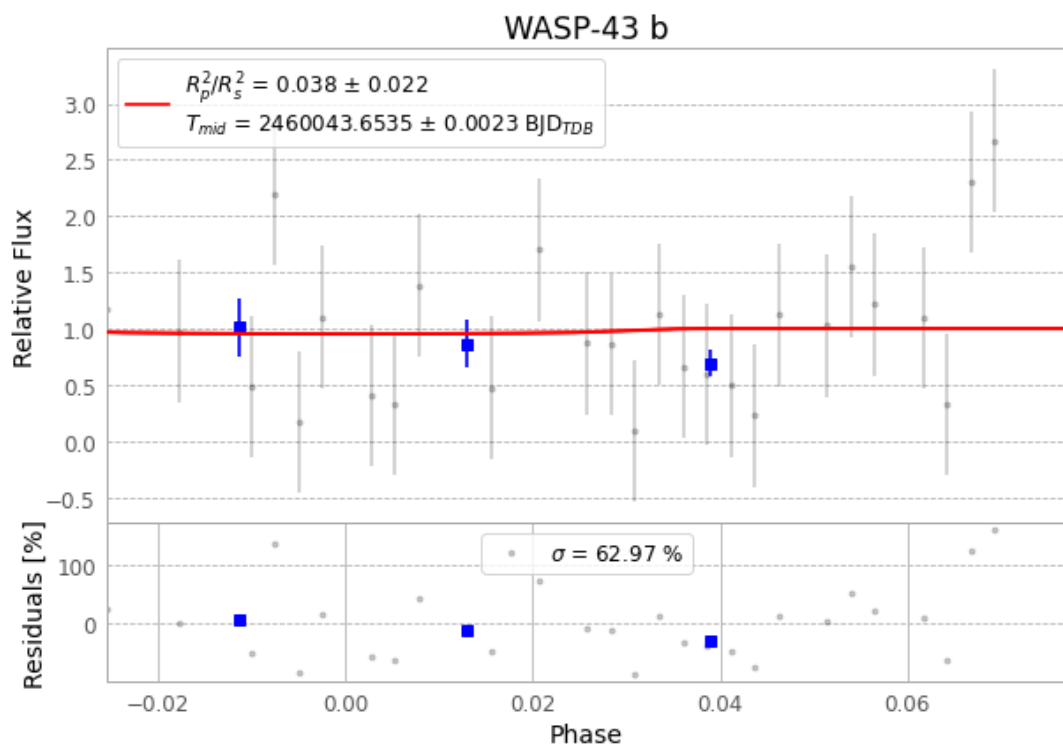


Figure 1 — FinalLightCurve_WASP-43 b_2023-04-08.png (SHA-256 c05b1abb9e338464...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [326, 221], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 850c829c230e878f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

39 FITS frames (26 MB), WASP-43230409030316.FITS → WASP-43230409050315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-230409.log (82bbd8df0b72...), FinalLightCurve_WASP-43 b_2023-04-08.png (c05b1abb9e33...), FinalLightCurve_WASP-43 b_2023-04-08.csv (bf634c036bfe...)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 09:42Z.